(d) Repeat **(c)** using the **larger** sphere.

d = ...

$1-\left(\frac{x}{d}\right)^2$ = ...

t = ...
[3]

(e) It is suggested that the relationship between t, h, x and d is

$$t^2 h = k\left(5+\frac{2}{1-\left(\frac{x}{d}\right)^2}\right)$$

where k is a constant.

(i) Using your data, calculate **two** values of k.

first value of k = ...

second value of k = ...
[1]

(ii) Justify the number of significant figures that you have given for your values of k.

....

....

.... [1]

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